THE RELATIVE PROJ OF A FINITELY PRESENTED GRADED ALGEBRA IS OF FINITE PRESENTATION WITH FINITELY PRESENTED TWISTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 7845-7853 and proof span 7855-7886 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof writes the graded presentation with the author's weights on the variables and degrees on the defining polynomials; both are preserved.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The compatibility of relative Proj with base change of the graded algebra, cited from the constructions chapter. Cited at proof lines 7869-7871. Lets the statement be checked over a finitely generated model ring and then transported back.
- The relative Proj of a finitely generated graded algebra is of finite type and its twists are quasi-coherent modules of finite type, cited from the same chapter. Cited at proof lines 7873-7876. Gives finite type for the model, which is upgraded to finite presentation over a Noetherian base.
- A finite type morphism of Noetherian schemes is of finite presentation, and a coherent module over a Noetherian scheme is of finite presentation, cited from the morphisms chapter and the cohomology of schemes chapter. Cited at proof lines 7878-7883. Upgrades finite type to finite presentation for both the morphism and the twists over the Noetherian model.

Source reference. Original lemma 01XZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2538–2538. Same-chapter reference used in the retained author text at source line(s) 166, 7884. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

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Source reference. Original lemma 01N2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2143–2143. Same-chapter reference used in the retained author text at source line(s) 7873, 8832. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 07ZY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10163–10163. Same-chapter reference used in the retained author text at source line(s) 7787, 7875. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01TX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2390–2390. Same-chapter reference used in the retained author text at source line(s) 7882. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{A}$ be a quasi-coherent graded $\mathcal{O}_S$-algebra. Let $p : X = \underline{Proj}_S(\mathcal{A}) \to S$ be the relative Proj of $\mathcal{A}$. If $\mathcal{A}$ is a finitely presented $\mathcal{O}_S$-algebra, then p is of finite presentation and $\mathcal{O}_X(d)$ is an $\mathcal{O}_X$-module of finite presentation.*

Proof. Affine locally this reduces to the following statement: Let R be a ring and let A be a finitely presented graded R-algebra. Then $\mathrm{Proj}(A) \to \mathrm{Spec}(R)$ is of finite presentation and $\mathcal{O}_{\mathrm{Proj}(A)}(d)$ is a $\mathcal{O}_{\mathrm{Proj}(A)}$-module of finite presentation. The finite presentation condition implies we can choose a presentation

$$A = R[X_1, \ldots, X_n]/(F_1, \ldots, F_m)$$

where $R[X_1, \ldots, X_n]$ is a polynomial ring graded by giving weights d_i to X_i and $F_1, \ldots, F_m$ are homogeneous polynomials of degree e_j. Let $R_0 \subset R$ be the subring generated by the coefficients of the polynomials $F_1, \ldots, F_m$. Then we set $A_0 = R_0[X_1, \ldots, X_n]/(F_1, \ldots, F_m)$. By construction $A = A_0 \otimes_{R_0} R$. Thus by Constructions, Lemma 01N2 it suffices to prove the result for $X_0 = \mathrm{Proj}(A_0)$ over R_0. By Lemma 07ZY we know X_0 is of finite type over R_0 and $\mathcal{O}_{X_0}(d)$ is a quasi-coherent $\mathcal{O}_{X_0}$-module of finite type. Since R_0 is Noetherian (as a finitely generated $\mathbf{Z}$-algebra) we see that X_0 is of finite presentation over R_0 (Morphisms, Lemma 01TX) and $\mathcal{O}_{X_0}(d)$ is of finite presentation by Cohomology of Schemes, Lemma 01XZ. This finishes the proof. □